STC: Inference-Time Semantic Target Control for Knowledge Graph Completion Rankings

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Abstract. Knowledge graph rankers are evaluated by held-out fact metrics, yet applications consume short top-k lists that may fail available type and relation checks. We formulate inference-time returned-list semantic control for a frozen ranker. STC operates on a finite top-M window and target quota q. Its exact query-specific solver, OptQ, certifies finite-window feasibility and computes the minimum query-specific pressure within the one-parameter controlled-score family needed to return at least q admissible candidates without retraining.
Across EurostatKG, DBpedia, and DrugBank with frozen PairRE, ComplEx, and RotatE rankers, OptQ at the canonical target $q = 5$ reduces Viol@10 by 0.183–0.488 while retaining Pres@10 = 0.512–0.823. Query-level 95% bootstrap intervals exclude zero for all semantic deltas; factual intervals are positive on EurostatKG and DrugBank and include zero on DBpedia. Fixed and learned policies expose a trade-off between insufficient and excessive control; quota and window sweeps show dataset-dependent feasibility–preservation frontiers. Isolated OptQ overhead is 1.02–4.71 ms/query. STC makes returned-list validity explicit and auditable, separately from retrieval and factual quality.

Keywords: Knowledge Graph Completion · Semantic Validation · Constrained Reranking · Inference-Time Control

1 Introduction

Knowledge graph completion (KGC) models are usually evaluated through held-out fact metrics such as MRR and Hits@k. These metrics remain essential, but do not characterize returned-list semantic validity [14]. A frozen ranker may expose candidates that fail available type or relation checks in its visible top-k, even when admissible alternatives already occur lower in the retrieved window. This failure mode is not tied to a particular scoring architecture [1,32,31,5].

Prior work incorporates semantic structure through type-aware scoring, rule guidance, logical regularization, and graph validation [19,10,11,35]. We study a later deployment stage: the scorer and a finite top-M candidate window are already materialized, and a downstream layer must return one bounded top-k

response. The controller neither retrains the model nor requests another window. Expanding retrieval is a complementary upstream strategy when the window contains too few admissible candidates; it does not address the within-window selection problem studied here.

The setting is also not equivalent to claiming that filtering is invalid. Hard validation is the strict endpoint when the available type and relation evidence is complete and every returned item must pass. In incomplete knowledge graphs, however, missing such evidence yields an unknown status rather than a confirmed violation, and an application may require a minimum admissibility quota while retaining as much of the learned order as possible. A single global penalty is softer than filtering but cannot adapt to query-specific score margins, semantic composition, and candidate availability.

We introduce Semantic Target Control (STC), an inference-time control framework for frozen KGC rankings. STC specifies the finite-window control problem, the operational semantic signal, and the distinction between retrieval and selection failure. Its exact query-specific solver, OptQ, certifies whether the top-M window contains at least q admissible candidates and, when feasible, computes the minimum query-specific semantic pressure within the one-parameter controlled-score family needed to meet the quota. Frozen scores and candidates are preserved, and deterministic tie-breaking makes the result auditable.

Experiments on EurostatKG, DBpedia, and DrugBank compare OptQ with the frozen ranking, quota filling, hard validation, tuned fixed penalties, and LightGBM rerankers. At $q = 5$, OptQ reduces Viol@10 by 0.183–0.488 while retaining Pres@10 = 0.512–0.823. We additionally report factual Hit@10/MRR@10 changes, query-level bootstrap intervals, energy and unknown-handling ablations, quota and window frontiers, cross-scorer and structural diagnostics, and verified runtime.

Our contributions are: (i) STC, a finite-window inference-time control framework that separates retrieval failure from correctable selection failure; (ii) OptQ, its exact query-specific solver, with a feasibility certificate and a minimum-pressure guarantee for a requested quota within the one-parameter controlled-score family; (iii) controlled comparisons with quota repair, strict filtering, fixed global penalties, and learned reranking; and (iv) a multi-dataset evaluation of semantic–factual trade-offs, statistical uncertainty, energy and window sensitivity, cross-scorer robustness, structural effects, and runtime.

2 Related Work

Knowledge graph completion is commonly formulated as ranking candidate entities for partially specified triples. Representative approaches include translational, bilinear, complex-valued, convolutional, rotational, and graph-neural models such as TransE, DistMult, ComplEx, ConvE, RotatE, PairRE, R-GCN, and CompGCN [1,39,32,6,31,5,29,33]. Although these architectures differ substantially, they are typically compared through held-out fact metrics such as MRR and Hits@k. Surveys and large empirical studies document the broader modeling, training, and evaluation landscape [25,13,26,17,27]. STC is complementary to this line: it does not introduce another scoring architecture or alter how factual rankings are learned, but controls the finite list exposed by an already trained scorer.

A second line incorporates semantic structure into KGC. Type-aware and rule-aware methods constrain candidates or training objectives [19,10,11,8], while SHACL validates graph data [35]. Sem@K measures relation-domain and relation-range compatibility in the first K predictions and shows that semantic and rank-based metrics are complementary [14]. Signature-aware losses use domain and range evidence during training [15]. These approaches modify the scorer, evaluate its ranking, or validate data. STC operates later: it freezes the scorer and candidate window, controls a requested admissibility quota in the visible list, and keeps missing evidence distinct from a confirmed violation.

Representative post-hoc KGC work spans general and relation-constrained reranking [37,22]; student, inductive, and text-enhanced pairwise rerankers [24,16,9]; and generative or retrieval–reasoning reranking [36,21]. LLM-based work validates predictions with contextual evidence [2], reranks top-k candidates with internal and external evidence [41], or addresses open-world KGC [12]; concurrent work studies ontology-constrained semantic reranking [42]. These approaches primarily optimize factual validation or rank quality. STC instead targets a visible admissibility quota under fixed upstream scores and a fixed candidate window, with finite-window feasibility and minimum-pressure certificates.

The returned-list perspective connects STC to information retrieval and ranking. Diversification methods such as maximal marginal relevance and xQuAD optimize visible-set properties beyond pointwise relevance [3,28]. Learning-to-rank methods likewise treat ordering as a learned decision problem [23]. Constrained-ranking work studies list-level exposure, representation, and fairness requirements [40,30,4]. These literatures show that aggregate scoring quality alone does not determine whether a returned list satisfies downstream requirements. Generic constrained-ranking methods, however, address representation or exposure requirements under different utility objectives.

STC lies at the intersection of semantic KGC and post-hoc returned-list control. It contributes a finite-window quota-control formulation that integrates operational semantic evidence with post-hoc returned-list control, together with OptQ, an exact query-specific solver. The framework separates retrieval failure from correctable selection failure, while OptQ computes the minimum pressure within the one-parameter controlled-score family. Hard validation remains the strict admissibility endpoint; fixed global penalties do not adapt to heterogeneous score margins and semantic compositions; and existing learned KGC rerankers target factual recovery rather than the same operational quota and certificate.

3 Problem Formulation

Let x denote a knowledge graph completion query, either a tail-prediction query $(h, r, ?)$ or a head-prediction query $(?, r, t)$. A frozen ranker assigns each candidate entity c a real-valued score $s_x(c)$. Sorting candidates by decreasing score induces the base ranking. We assume that the downstream controller can access only the finite candidate window

$$C_M(x) = \mathrm{Top}_M(s_x)\,,$$

with deterministic tie-breaking. The list exposed to a user, API, or downstream component contains only k candidates, where $k \ll M$. We denote the frozen visible list by

$$L_k^0(x) = \mathrm{Top}_k(C_M(x); s_x)\,.$$

Operational assumptions and terminology. The scorer s_x, materialized top-M window $C_M(x)$, visible budget k, and status assignment remain fixed; the window is not re-queried or expanded. *Feasibility* means $|A_x| \geq q$. *Controllability* means that some $\lambda \geq 0$ in the scalar family $s_x - \lambda e_x$ returns a top-k list meeting the quota. *Semantic pressure* is λ, and *semantic energy* is e_x. Candidate-level *admissibility* is operational: a *violation* is a failure of the stated candidate-level test, not an OWL open-world inconsistency claim or a factual-error judgment.

For each $c \in C_M(x)$, an external validation process assigns an operational semantic status

$$z_x(c) \in \{\mathsf{adm}, \mathsf{viol}, \mathsf{unk}\}\,.$$

The status is computed from the semantic evidence available at inference time, including processed type assertions, relation-domain and relation-range constraints, and disjointness-derived incompatibilities. A candidate is adm when it passes all applicable checks, viol when it fails at least one applicable check, and unk when the available evidence is insufficient to establish either outcome. Missing type evidence yields unk, not viol.

Let

$$A_x = \{c \in C_M(x) : z_x(c) = \mathsf{adm}\}$$

be the admissible candidates in the available window. For any returned list $L \subseteq C_M(x)$, define its admissible count as

$$\mathrm{Adm}_x(L) = \sum_{c \in L} \mathbf{1}[z_x(c) = \mathsf{adm}]\,.$$

Unknown candidates do not contribute to the requested admissibility quota, although their treatment can be adjusted through the control energy defined below. *Returned-list semantic validity* is the list-level property summarized by admissible count, violation rate, or quota satisfaction; it is distinct from factual correctness.

Each candidate is also assigned a non-negative semantic energy $e_x(c)$. In the main configuration, admissible candidates receive zero energy, violating candidates receive positive energy, and unknown candidates receive a configurable non-negative penalty. More generally, the energy can encode the strength or combination of the applicable semantic checks. It is fixed at inference time, is not learned jointly with the ranker, and does not alter the parameters of the frozen scoring model.

Given a visible-list size k and a requested quota $q \in \{0, \dots, k\}$, the returned-list control problem is to produce a list $L_k(x) \subseteq C_M(x)$ satisfying

$$|L_k(x)| = k \qquad \text{and} \qquad \mathrm{Adm}_x(L_k(x)) \geq q,$$

while intervening on the frozen ranking as little as possible. The controller cannot introduce candidates outside $C_M(x)$. In Section 4, minimal intervention is made precise through a one-parameter controlled-score family and the smallest non-negative pressure that satisfies the quota.

The quota is feasible within the available candidate window exactly when

$$\mathcal{F}_q(x) = \mathbf{1}[|A_x| \geq q] = 1.$$

This certificate concerns the observed top-M window, not the full entity universe. If $|A_x| < q$, no reranker restricted to $C_M(x)$ can return q admissible candidates; the target is finite-window infeasible, which constitutes a *retrieval failure*. If $|A_x| \geq q$ but

$$\mathrm{Adm}_x\big(L_k^0(x)\big) < q,$$

the required candidates are available but are not exposed by the frozen top-k; this is a *selection failure*. No intervention is needed when the frozen list already satisfies the quota. STC addresses selection failure and explicitly certifies retrieval failure rather than claiming to recover candidates that were never retrieved.

4 Semantic Target Control

Semantic Target Control (STC) operates after a frozen ranker has produced the candidate window $C_M(x)$ and before the visible top-k list is returned. It consumes the frozen scores, the operational semantic status and energy of each candidate, and the requested admissibility quota q, which defines the semantic target for the returned top-k list. It neither retrains the ranker nor modifies its parameters or candidate generator. Figure 1 summarizes this fixed-window inference-time control pipeline. STC first checks whether the quota is feasible within $C_M(x)$. If the window contains fewer than q admissible candidates, it reports a retrieval failure. Otherwise, the frozen top-k is returned unchanged if it already satisfies the quota; if not, OptQ applies the minimum query-specific pressure needed to produce a quota-satisfying list.

4.1 OptQ: Query-specific finite-window control

For a query x and candidate $c \in C_M(x)$, STC defines the controlled score

$$s_{x,\lambda}(c) = s_x(c) - \lambda e_x(c), \qquad \lambda \geq 0, \tag{1}$$

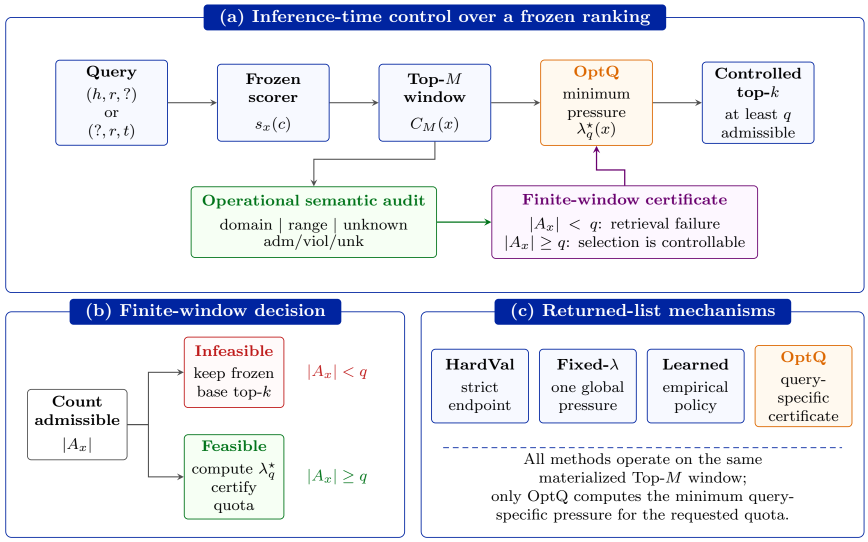

Fig. 1. Overview of the STC finite-window control pipeline over a frozen KGC ranking.

where $s_x(c)$ is the frozen score and $e_x(c) \geq 0$ is the semantic energy. Setting $\lambda = 0$ recovers the frozen ranking. Increasing λ lowers the scores of candidates with positive energy while leaving zero-energy admissible candidates unchanged.

All controlled rankings use a fixed lexicographic ordering

$$\kappa_{x,\lambda}(c) = \Big(s_{x,\lambda}(c), -e_x(c), s_x(c), -\pi_x(c)\Big), \tag{2}$$

sorted in decreasing order, where $\pi_x(c)$ is the original position of c in $C_M(x)$. Thus, equal controlled scores are resolved by lower semantic energy, higher frozen score, and finally the original candidate order. This convention makes quota satisfaction at score–energy crossings deterministic.

The exact controller used in the main experiments assumes the zero-versus-positive energy structure

$$e_x(c) = 0 \quad \text{for} \quad z_x(c) = \mathsf{adm}, \qquad e_x(c) > 0 \quad \text{otherwise}.$$

The positive energies need not be equal. Unknown-aware configurations satisfy the same condition by assigning unknown candidates a strictly positive configurable penalty.

Let $A_x = \{c \in C_M(x) : z_x(c) = \mathsf{adm}\}$ denote the admissible candidates in the available window. For $q = 0$, no intervention is required. For $q \geq 1$, if $|A_x| < q$, the quota is infeasible within the candidate window. Otherwise, let a_q be the q-th highest-ranked candidate in A_x under the frozen ordering and define

$$\tau_q = s_x(a_q).$$

Consequently, at most $b = k - q$ positive-energy candidates may remain above a_q while still allowing the top-k list to contain q admissible candidates.

For every positive-energy candidate initially above this threshold, define its crossing pressure

$$\gamma_x(c) = \frac{s_x(c) - \tau_q}{e_x(c)}, \qquad c \notin A_x, \quad s_x(c) > \tau_q. \tag{3}$$

Let $\gamma_{(1)} \geq \gamma_{(2)} \geq \cdots \geq \gamma_{(m)}$ denote these pressures in non-increasing order. OptQ returns

$$\lambda_q^\star(x) = \begin{cases} 0, & m \leq k - q, \\ \gamma_{(k-q+1)}, & m > k - q. \end{cases} \tag{4}$$

The controlled returned list is then

$$L_k^\star(x) = \mathrm{Top}_k \left(C_M(x); \kappa_{x,\lambda_q^\star(x)} \right). \tag{5}$$

Equation (4) has a direct interpretation. OptQ does not force every positive-energy candidate below every admissible candidate. It permits the $k-q$ positions not occupied by admissible candidates, as allowed by the requested quota and applies only the pressure needed to remove the next positive-energy candidate from above the q-th admissible threshold. Minimality therefore refers to the smallest scalar pressure within the controlled-score family in Equation (1); it is not a claim of globally minimum permutation distance over arbitrary reranking procedures. Extended derivations, tie cases, and the $q = k$ strict endpoint appear in Supplementary Annex B.

Scope of the guarantee. The quota guarantee is conditional on the specified operational energy and the candidates available in $C_M(x)$. It guarantees the requested zero-energy admissible count under that signal; it does not assert signal completeness, factual truth, or global OWL consistency. Minimality is restricted to the scalar controlled-score family in Equation (1), not to arbitrary permutations or methods that retrieve additional candidates. Algorithm 1 gives the complete OptQ procedure.

4.2 Guarantee and deployment properties

The following proposition states the finite-window guarantee underlying Algorithm 1.

Proposition 1. Fix a query x, a finite candidate window $C_M(x)$, integers $1 \leq q \leq k \leq M$, and the deterministic ordering in Equation (2). Assume that admissible candidates have zero energy and all other candidates have strictly positive energy.

If $|A_x| < q$, no returned list of size k drawn from $C_M(x)$ can satisfy the quota. If $|A_x| \geq q$, the list $L_k^\star(x)$ produced by OptQ contains at least q admissible candidates. Moreover, for every $0 \leq \lambda < \lambda_q^\star(x)$, the controlled top-$k$ list under $\kappa_{x,\lambda}$ contains fewer than q admissible candidates.

Proof sketch. When $|A_x| < q$, the candidate window itself contains too few admissible candidates, so no reranking restricted to that window can satisfy the

Algorithm 1 OptQ finite-window quota control

Require: Window $C_M(x)$, scores s_x, statuses z_x, energies e_x, visible size k, quota q, with $0 \leq q \leq k \leq M$
Assumptions: $e_x(c) = 0$ if $z_x(c) = \mathsf{adm}$, $e_x(c) > 0$ otherwise; ordering $\kappa_{x,\lambda}$ from Eq. (2)
Ensure: Feasibility flag; if feasible, minimum $\lambda_q^\star(x)$ and quota-satisfying $L_k^\star(x)$; otherwise null pressure and frozen $L_k^0(x)$

if $q = 0$ **then**
  return feasible, 0, $\mathrm{Top}_k(C_M(x); \kappa_{x,0})$
end if
$A_x \leftarrow \{c \in C_M(x) : z_x(c) = \mathsf{adm}\}$
if $|A_x| < q$ **then**
  return infeasible, null, $\mathrm{Top}_k(C_M(x); \kappa_{x,0})$
end if
$a_q \leftarrow q$-th candidate in A_x under $\kappa_{x,0}$
$\tau_q \leftarrow s_x(a_q)$
$b \leftarrow k - q$
$\Gamma \leftarrow [\,]$
for $c \in C_M(x) \setminus A_x$ **do**
  if $e_x(c) > 0$ and $s_x(c) > \tau_q$ **then**
    append $(s_x(c) - \tau_q)/e_x(c)$ to Γ
  end if
end for
sort Γ in non-increasing order
if $|\Gamma| \leq b$ **then**
  $\lambda_q^\star(x) \leftarrow 0$
else
  $\lambda_q^\star(x) \leftarrow \Gamma_{b+1}$
end if
$L_k^\star(x) \leftarrow \mathrm{Top}_k(C_M(x); \kappa_{x,\lambda_q^\star(x)})$
return feasible, $\lambda_q^\star(x)$, $L_k^\star(x)$

target. Now assume $|A_x| \geq q$, and consider the q-th admissible candidate a_q with score τ_q. A positive-energy candidate c remains strictly above a_q exactly while

$$\lambda < \frac{s_x(c) - \tau_q}{e_x(c)}.$$

At $\lambda = \lambda_q^\star(x)$, at most $k - q$ such candidates remain strictly above a_q; ties at the threshold are resolved in favor of the lower-energy admissible candidate. Hence all q highest-scoring admissible candidates fit in the visible top-k. For any smaller pressure, at least $k - q + 1$ positive-energy candidates remain above a_q, leaving at most $q - 1$ positions for those admissible candidates. Therefore the quota is not yet satisfied.

The guarantee is explicitly finite-window. If the requested quota is infeasible, STC reports the failure and deployment-style evaluation returns the frozen top-k list rather than applying an arbitrary unsuccessful intervention. A feasible query whose frozen list already meets the quota has $\lambda_q^\star(x) = 0$, so STC leaves it unchanged. Computationally, the closed form avoids enumerating all pairwise score–energy crossings: only crossings with the q-th admissible threshold are required. After scores, statuses, and energies are available, constructing the crossing set is $O(M)$. The reference implementation sorts the crossings and therefore requires $O(M \log M)$ time and $O(M)$ additional memory; selecting only the required order statistic can reduce the pressure computation to linear expected time. Constructing the final top-k list requires $O(M \log k)$ time with a bounded heap. The controller is therefore a post-scoring operation whose measured overhead is reported separately in Section 6.

5 Experimental Setup

We evaluate STC as a returned-list control layer over frozen knowledge graph rankers. All policies consume precomputed frozen scores and operate only within the corresponding top-M candidate window; none retrieves additional candidates, retrains the upstream model, or changes its parameters.

The evaluation covers EurostatKG [34], DBpedia [20], and DrugBank [38]. EurostatKG provides a controlled administrative and statistical setting with substantial factual-target exposure. DBpedia is evaluated through the canonical paper subset and represents an incomplete Web knowledge graph in which missing type evidence makes unknown handling particularly important. DrugBank provides a large biomedical stress test with strong semantic correction pressure; its relation concentration is analyzed separately rather than treated as evidence of broad relation diversity.

We designate PairRE for EurostatKG, ComplEx for DBpedia, and RotatE for DrugBank as the canonical configurations used in the main analysis. Beyond these configurations, we conduct three additional full-scope runs: EurostatKG with ComplEx, DrugBank with ComplEx, and DrugBank with PairRE. These runs test whether the observed semantic effect depends on a single embedding architecture. The cross-scorer analysis therefore covers EurostatKG and DrugBank. The evaluated entity universes contain 44,948, 5,496,987, and 22,466 entities for EurostatKG, DBpedia, and DrugBank, respectively. Table 1 summarizes the evaluation populations together with finite-window feasibility, factual-target exposure, and the completeness of the available typing evidence.

Table 1. Canonical evaluation and semantic-evidence diagnostics at $q = 5$. Feas. is finite-window quota feasibility; Target@M is factual-target exposure; Type cov. is dataset-wide entity typing coverage after preprocessing; and Unk@10 is the rounded unknown share after control.

Dataset	Queries	M	Feas.	Target@M	Type cov.	Unk@10
EurostatKG	12,922	20,000	0.780	0.543	0.995	0.002
DBpedia	47,246	5,000	0.687	0.0014	0.842	0.213
DrugBank	585,952	5,000	1.000	0.571	1.000	0.000

The diagnostics expose distinct limitations. DBpedia combines incomplete typing evidence with negligible factual-target exposure, whereas DrugBank has unit typing coverage under its induced preprocessing pipeline and is quota-feasible at the canonical window. Accordingly, semantic, retrieval, and factual outcomes are analyzed separately throughout the evaluation.

The main deployment configuration uses $k = 10$, $q = 5$, and $M = 20{,}000$ for EurostatKG or $M = 5{,}000$ for DBpedia and DrugBank. Candidate status is computed from the processed type closure and the applicable relation-domain and relation-range constraints; no general-purpose OWL reasoner is used. A violating status records a failed check, whereas unknown denotes insufficient evidence rather than violation. The canonical energy assigns zero energy to admissible candidates and positive energy to violating and unknown candidates, with a unit unknown penalty. Energy ablations separate the domain and range signals and vary the treatment of unknown candidates.

We evaluate eight returned-list policies on the same frozen candidate windows. **Base** is the unmodified frozen ranking. **QuotaFill** minimally replaces visible items not classified as admissible with admissible candidates from the same window until quota q is met. **HardVal** prioritizes candidates not confirmed as violations under the available checks and retains unknown candidates; if fewer than k such candidates remain, it fills the list from the original ranking. **Fixed-nearPres** and **Fixed-safe** use one globally selected pressure for every query, chosen respectively to approach OptQ preservation and to reach quota success 1.0. **Learned-nearPres** and **Learned-safe** are two diagnostic operating points from a LightGBM candidate reranker [18]. They represent preservation-oriented and quota-oriented learned settings rather than a new upstream KGC model. **OptQ** is the proposed query-specific controller. These policies directly test strict filtering, score-preserving quota repair, one tuned global penalty, and globally trained learned reranking.

The LightGBM classifier is fitted on training-split queries, its mixing-weight grid is evaluated on validation-split queries, and final metrics use held-out evaluation queries; full features, hyperparameters, split sizes, and validation summaries are provided in the supplementary material.

We use two complementary evaluation protocols. The *all-query deployment protocol* evaluates the canonical test population and returns the frozen top-k list whenever the requested quota is infeasible in the available candidate window. This protocol is used for the main factual-utility, candidate-window, and bootstrap results. The *matched selection-failure protocol* evaluates only feasible queries whose frozen top-k contains fewer than q admissible candidates. Learned-policy and fixed-pressure experiments use separate held-out populations, whose counts are given in their captions. Every table is internally matched; its OptQ row is a table-specific reference and is not compared across populations.

Candidate-window sensitivity fixes $q = 5$ and varies

$$M \in \{500, 1000, 5000, 10000, 20000\}$$

for EurostatKG and

$$M \in \{500, 1000, 2500, 5000\}$$

for DBpedia and DrugBank. Every point uses the same canonical query population and deployment fallback as the corresponding main configuration. Quota sensitivity instead fixes the canonical candidate window and varies q, thereby separating the effect of candidate availability from the requested degree of semantic intervention.

Table 1 reports the rounded controlled-list shares for the canonical configurations together with dataset-wide typing coverage recorded in the preprocessing `preflight.json` files. The energy ablations separately vary the domain, range, and unknown-penalty components.

We report complementary semantic, ranking, factual, and computational metrics. Adm@k and Unk@k are returned-list shares. Viol@k is the fraction of confirmed violations among returned candidates for which the operational checks are evaluable; therefore, the three quantities need not sum to one. Quota success records whether the returned list contains at least q admissible candidates, while feasibility records whether the candidate window contains enough admissible candidates for that target to be attainable. Pres@k measures overlap with the frozen top-k, and Shift@k summarizes the rank movement introduced by the controller. Hit@10 and MRR@10 measure truncated factual utility, and runtime is decomposed to isolate the OptQ control overhead from candidate scoring and semantic status computation.

Statistical uncertainty is estimated with 1,000 query-level bootstrap resamples [7] over the complete canonical evaluation populations: 12,922 queries for EurostatKG, 47,246 for the DBpedia paper subset, and 585,952 for DrugBank. The bootstrap point estimates match the corresponding all-query main results and evaluation counts. Supplementary Annexes A and C provide dataset, model, configuration, and reproducibility details.

6 Results

Using the protocol above, we first report the canonical all-query results, then compare OptQ with direct and learned alternatives, and finally examine fixed-pressure baselines, energy ablations, bootstrap uncertainty, and runtime.

Table 2 reports the canonical $q = 5$ all-query results. OptQ reduces Viol@10 by 0.269, 0.183, and 0.488 on EurostatKG, DBpedia, and DrugBank while retaining Pres@10 = 0.731, 0.823, and 0.512. The factual deltas are positive on EurostatKG and DrugBank. DBpedia has extremely low factual-target exposure, so its near-zero factual changes are reported transparently but not interpreted as an improvement.

Unknown candidates remain distinct from confirmed violations. Their rounded shares after control are 0.002, 0.213, and 0.000 for EurostatKG, DBpedia, and DrugBank. The larger DBpedia share is consistent with incomplete type evidence and motivates the explicit unknown-aware energy analysis below.

Table 3 compares OptQ with direct alternatives on the held-out sampled matched populations. QuotaFill reaches the target with high preservation but is a discrete replacement policy rather than a minimum-pressure score controller.

Table 2. Canonical all-query OptQ results at $q = 5$. Deltas are measured against the frozen ranking; exact confidence intervals are reported in Table 6.

Dataset	ΔHit	ΔMRR	ΔViol	ΔAdm	Pres.	Shift
EurostatKG	+0.0065	+0.0018	-0.269	+0.269	0.731	129.8
DBpedia	+0.000042	+0.000004	-0.183	+0.177	0.823	49.4
DrugBank	+0.0012	+0.0002	-0.488	+0.488	0.512	425.6

Table 3. Held-out sampled $q = 5$ policy comparison on feasible selection-failure queries (734/570/9,891 queries for EurostatKG/DBpedia/DrugBank). Q denotes quota success.

Dataset	Policy	Q	Viol.	Adm.	Pres.
EurostatKG	QuotaFill	1.000	0.492	0.508	0.620
	HardVal	0.985	0.000	0.987	0.128
	Learned-nearPres	0.711	0.384	0.616	0.456
	Learned-safe	1.000	0.000	1.000	0.072
	OptQ	1.000	0.497	0.503	0.569
DBpedia	QuotaFill	1.000	0.480	0.506	0.650
	HardVal	0.685	0.000	0.684	0.200
	Learned-nearPres	0.725	0.383	0.610	0.531
	Learned-safe	1.000	0.067	0.931	0.210
	OptQ	1.000	0.518	0.468	0.673
DrugBank	QuotaFill	1.000	0.500	0.500	0.500
	HardVal	1.000	0.000	1.000	0.000
	Learned-nearPres	0.212	0.772	0.228	0.773
	Learned-safe	1.000	0.000	1.000	0.000
	OptQ	1.000	0.500	0.500	0.500

Learned-nearPres does not reliably meet the quota, with quota success 0.711, 0.725, and 0.212, whereas Learned-safe reaches the target by sharply reducing preservation. OptQ reaches quota success 1.0 on all three populations and retains 0.569, 0.673, and 0.500 of the frozen top-10.

A single global semantic pressure exhibits the same trade-off between insufficient and excessive control on the full matched populations. In Table 4, the Near settings satisfy only 0.489–0.587 of matched queries. The Safe settings attain quota success 1.0, but preservation falls to 0.000–0.166, compared with 0.500–0.650 for query-specific OptQ. Because the learned and fixed-pressure experiments use different held-out populations, their OptQ reference rows establish within-table comparability and are not cross-table replicates.

Table 5 tests dependence on the operational semantic signal. Domain-only and range-only checks produce different feasibility and intervention profiles, while the unknown penalty changes the balance between violation and unknown statuses. The combined domain–range configuration is therefore a documented operating point rather than an assumed application-independent truth. DrugBank has zero

Table 4. Query-specific OptQ versus globally fixed pressure on the full matched selection-failure populations (9,130/25,889/571,893 queries). Near is preservation-oriented; Safe attains quota success 1.0.

	OptQ	Near fixed-λ			Safe fixed-λ	
Dataset	Pres.	λ	Q	Pres.	λ	Pres.
EurostatKG	0.620	0.1	0.587	0.567	3	0.128
DBpedia	0.650	3	0.489	0.704	100	0.166
DrugBank	0.500	3	0.534	0.462	30	0.000

Table 5. Operational-energy ablation. D+R combines domain and range checks; u is the unknown-candidate penalty.

Dataset	Energy	Feas.	ΔViol	ΔAdm	ΔUnk	Pres.
EurostatKG	D+R, $u = 1$	0.780	-0.269	+0.269	0.000	0.731
	Domain	0.542	-0.069	+0.069	0.000	0.931
	Range	0.993	-0.171	+0.170	-0.002	0.830
	D+R, $u = 0.5$	0.780	-0.269	+0.269	+0.001	0.730
	D+R, $u = 2$	0.780	-0.269	+0.269	0.000	0.731
DBpedia	D+R, $u = 1$	0.687	-0.183	+0.177	-0.010	0.823
	Domain	0.452	-0.095	+0.090	-0.010	0.910
	Range	0.554	-0.130	+0.123	-0.014	0.877
	D+R, $u = 0.5$	0.687	-0.224	+0.177	+0.036	0.787
	D+R, $u = 2$	0.687	-0.176	+0.177	-0.021	0.823

unknown share under the induced typing pipeline, so changing the unknown penalty has no effect and those variants are not repeated.

Supplementary Annex D reports complete quota/window frontiers, cross-scorer results, structural slices, and query-level diagnostics. Their main conclusion is consistent across datasets: stricter quotas improve admissibility at greater preservation cost, while larger M increases feasibility. At $q = k$, canonical binary-energy OptQ coincides with admissibility-first ranking on feasible queries; the equal-positive-energy proof is supplementary. EurostatKG saturates by $M = 10{,}000$, DBpedia increases gradually to 0.687, and DrugBank rises from 0.195 at $M = 500$ to approximately 1.0 at $M = 5{,}000$.

Three additional scorer runs on EurostatKG and DrugBank preserve the semantic-effect direction ($\Delta\text{Viol@10} = -0.276, -0.179, -0.124$); complete cross-scorer, relation, and head–tail diagnostics are supplementary. DrugBank remains a relation-skewed stress test. Consistently, the bootstrap intervals in Table 6 exclude zero for every canonical semantic delta. EurostatKG and DrugBank also have strictly positive factual intervals. Both DBpedia factual intervals begin at zero, supporting no claim of material factual change.

Finally, Table 7 isolates the inference-time control cost. OptQ adds 4.71, 1.02, and 1.34 ms/query on EurostatKG, DBpedia, and DrugBank, respectively; candidate construction and semantic status evaluation remain the dominant costs.

Table 6. Query-level 95% bootstrap intervals for canonical OptQ deltas (1,000 resamples).

Dataset	ΔHit@10	ΔMRR@10
EurostatKG	+0.006501 [+0.005185, +0.007894]	+0.001817 [+0.001357, +0.002335]
DBpedia	+0.000042 [+0.000000, +0.000106]	+0.000004 [+0.000000, +0.000011]
DrugBank	+0.001164 [+0.001084, +0.001249]	+0.000206 [+0.000187, +0.000225]
Dataset	ΔViol@10	ΔAdm@10
EurostatKG	−0.2686 [−0.2726, −0.2650]	+0.2686 [+0.2649, +0.2726]
DBpedia	−0.1826 [−0.1846, −0.1806]	+0.1769 [+0.1751, +0.1787]
DrugBank	−0.4879 [−0.4881, −0.4877]	+0.4879 [+0.4877, +0.4881]

Table 7. Verified runtime decomposition on representative samples. Loop throughput is queries/s; semantic checking and OptQ are ms/query.

Dataset	M	Loop q/s	Semantic ms/q	OptQ ms/q
EurostatKG	20,000	38.7	17.68	4.71
DBpedia	5,000	7.0	6.24	1.02
DrugBank	5,000	146.9	3.53	1.34

7 Discussion and Limitations

7.1 Interpretation and deployment

Taken together, the results confirm STC's finite-window interpretation: infeasible windows are reported as retrieval failures, compliant lists remain unchanged, and OptQ intervenes only for feasible selection failures. Under the stated zero-versus-positive energies and deterministic tie-breaking, it applies the smallest non-negative pressure in the controlled-score family that meets the quota. The guarantee is query-specific and conditional on the operational semantic signal.

The semantic and factual results answer different questions. Reductions in returned-list violations show that the operational domain, range, and type checks exert the intended influence on the visible list; they do not establish that the promoted candidates are factually correct answers. Factual changes also require the held-out target to occur in the candidate window and to be affected by reranking. EurostatKG and DrugBank show small positive factual effects, whereas DBpedia shows no material factual change despite a clear semantic improvement. Returned-list semantic metrics and factual ranking metrics should therefore be reported and interpreted separately.

The quota q and window size M govern different deployment axes. Increasing q requests stronger admissibility control and generally reduces preservation. Increasing M expands the available opportunity set and can improve feasibility, but also raises candidate construction and semantic status evaluation costs. A deployment can therefore choose M under latency and memory constraints and select q on held-out validation data from the resulting semantic–preservation frontier. The observed sweeps illustrate the dataset dependence of this choice: EurostatKG reaches an early plateau, DBpedia improves gradually, and DrugBank has a pronounced small-window retrieval bottleneck.

The baseline comparisons clarify the scope of query-specific control.

QuotaFill performs discrete replacement, while HardVal prioritizes candidates not confirmed as violations, retains unknowns, and uses the original ranking as a size-preserving fallback; global fixed or learned settings trade preservation against quota failures. OptQ addresses the narrower objective of meeting a visible per-query quota with minimum scalar pressure in its defined score family.

STC is therefore most appropriate when the scorer and candidate window are fixed, a downstream consumer observes only a bounded list, and operational semantic admissibility matters without requiring complete filtering. When the semantic evidence is trusted and only admissible outputs are acceptable, hard validation remains appropriate. When metadata are very sparse, the formal certificate remains relative to those labels, but its substantive value shifts toward diagnosing whether failure originates in candidate availability or in the ordering of available candidates.

7.2 Limitations and scope

The principal limitation is the quality and specification of the operational semantic signal. Candidate statuses are derived from processed type, domain, and range information and can inherit missing assertions, schema errors, and preprocessing choices. Treating missing evidence as unknown avoids equating it with a confirmed violation, but does not recover the candidate's true status. Moreover, the energy function encodes a control policy rather than an application-independent notion of semantic truth. The guarantees consequently hold only with respect to the supplied statuses and energies. The ablations document sensitivity to domain, range, and unknown handling; they do not establish a universally optimal semantic signal. DrugBank's zero unknown share under the induced typing pipeline means that changing the unknown penalty has no effect.

STC also cannot repair upstream retrieval. Its feasible set is bounded by $C_M(x)$, and increasing M may be expensive for large candidate universes. The candidate-window experiments quantify this dependence but do not remove it. A different candidate generator could change both quota feasibility and the preservation frontier, so joint retrieval and returned-list control remains a separate problem.

The empirical evidence covers three knowledge graphs and a limited set of frozen embedding-model configurations. EurostatKG and DBpedia retain relatively broad relation coverage in the evaluated subsets, whereas DrugBank is strongly relation-concentrated and is best interpreted as a large-scale correction-pressure stress test. Additional domains, candidate-generation regimes, text-augmented rankers, and production ranking services are needed to establish the transferability of the observed trade-offs.

The closed-form guarantee further assumes zero energy for admissible candidates, strictly positive energy for every other candidate, and the stated deterministic tie-breaking rule. Alternative energy constructions may still be useful, but their formal properties must be established separately. Learned or language-model-based validators could supply semantic signals when structured metadata are unavailable, but their calibration and uncertainty would then become part of the control problem. Any guarantee would remain relative to the statuses and energies induced by those validators.

Finally, the abstraction of a frozen ranker, an external constraint signal, a finite window, and a visible quota may extend beyond knowledge graph completion, but the present experiments establish only KG-side operational admissibility. Claims concerning safety, fairness, coverage, or other policy-constrained rankings require domain-specific definitions and separate evaluation. STC does not improve the factual ranker, complete missing semantic metadata, or repair an ontology. Within these boundaries, it addresses a specific deployment problem: controlling how fixed semantic evidence influences a bounded list after ranking.

8 Conclusion

This work introduced Semantic Target Control (STC), an inference-time control layer for frozen knowledge graph rankers. Given a finite candidate window and a requested admissibility quota, STC separates retrieval failure from correctable selection failure and uses OptQ to compute the minimum query-specific pressure within the controlled score family that satisfies the target. The upstream scorer and candidate generator remain unchanged.

Operationally, infeasible windows require retrieval expansion; feasible quota misses are selection failures corrected by minimum pressure; and already compliant lists remain unchanged. Hard validation remains appropriate when every returned item must be admissible, while sparse metadata limit guarantees to the observed admissible, violating, and unknown statuses.

Across three knowledge graphs and multiple frozen rankers, OptQ improves returned-list semantic validity while making preservation cost explicit. Quota and window analyses characterize finite-window behavior, bootstrap intervals quantify uncertainty, and runtime measurements establish practical cost; alternative policies lack the same query-specific certificate and minimum-pressure property.

STC makes returned-list semantic validity explicit, auditable, and controllable without replacing factual ranking, retrieval, or validation. Future work may add adaptive retrieval or learned semantic signals while preserving the distinction among evidence, finite-window feasibility, and intervention.

Supplementary Material Statement: Supplementary Annexes A–D, available in the linked supplementary material, provide extended proofs, dataset and protocol details, robustness analyses, and reproducibility information. The GitHub repository contains the code, configurations, reproducibility metadata, provenance records, and final publication files. Redistributable large artifacts are available in the external artifact folder. For resources that cannot be redistributed, the repository provides reconstruction guidance and scripts, subject to the applicable data-access terms.

Declaration of Use of Generative AI OpenAI ChatGPT was used to assist with language editing, manuscript organization, development and debugging of research code and automation scripts, and presentation of experimental analyses. All outputs were reviewed and validated by the authors, who take full responsibility for the manuscript.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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